

Sixdof Falcon
Build Examples

Version 6

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1 Introduction

The goal of this document is to explain how to run the build examples included with the SdsFalcon SDK.

The code for the build examples sits in the `examples` folder.

The goal of the build examples is to ensure that users have their environment properly configured and can seamlessly connect to the sensor, enabling them to receive accurate output data.

2 Setting Up Visual Studio Code

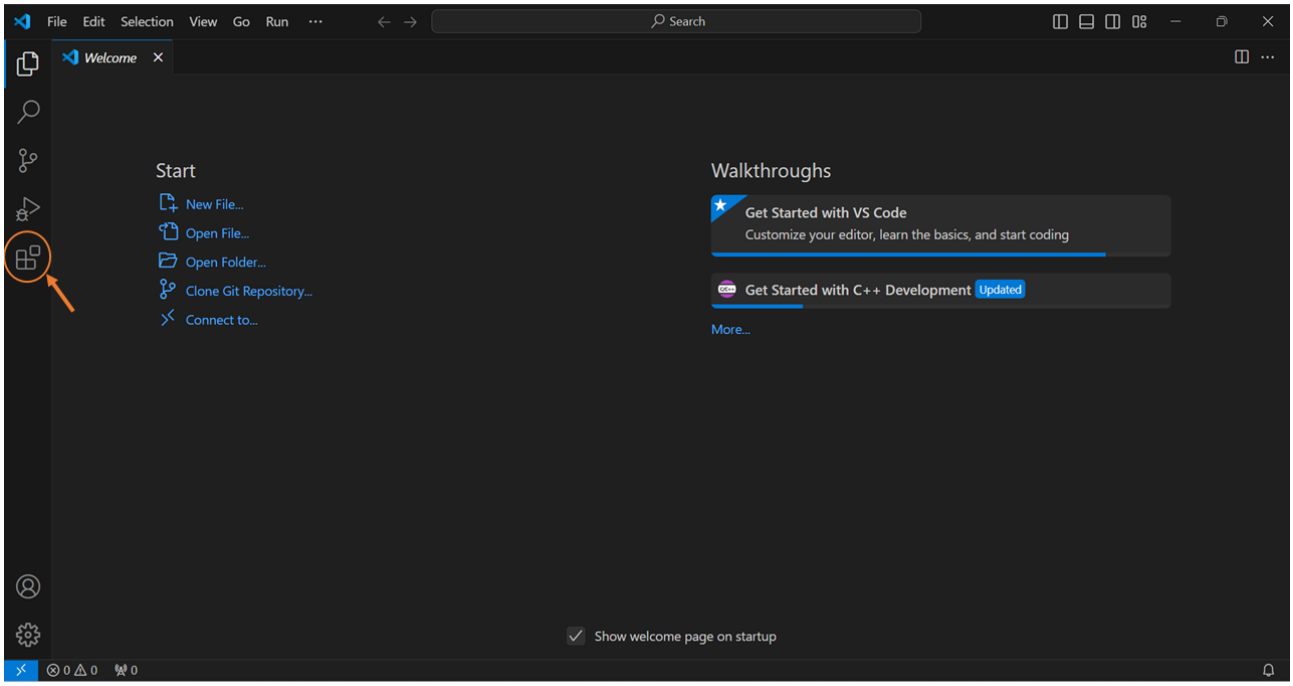
In this section, we will guide you through each step of setting up Visual Studio Code on your computer. If you have already completed this setup, please ensure that you have all the required extensions installed before proceeding to the next section.

2.1 Installation

To begin, download and install Visual Studio Code from the official website: Visual Studio Code.

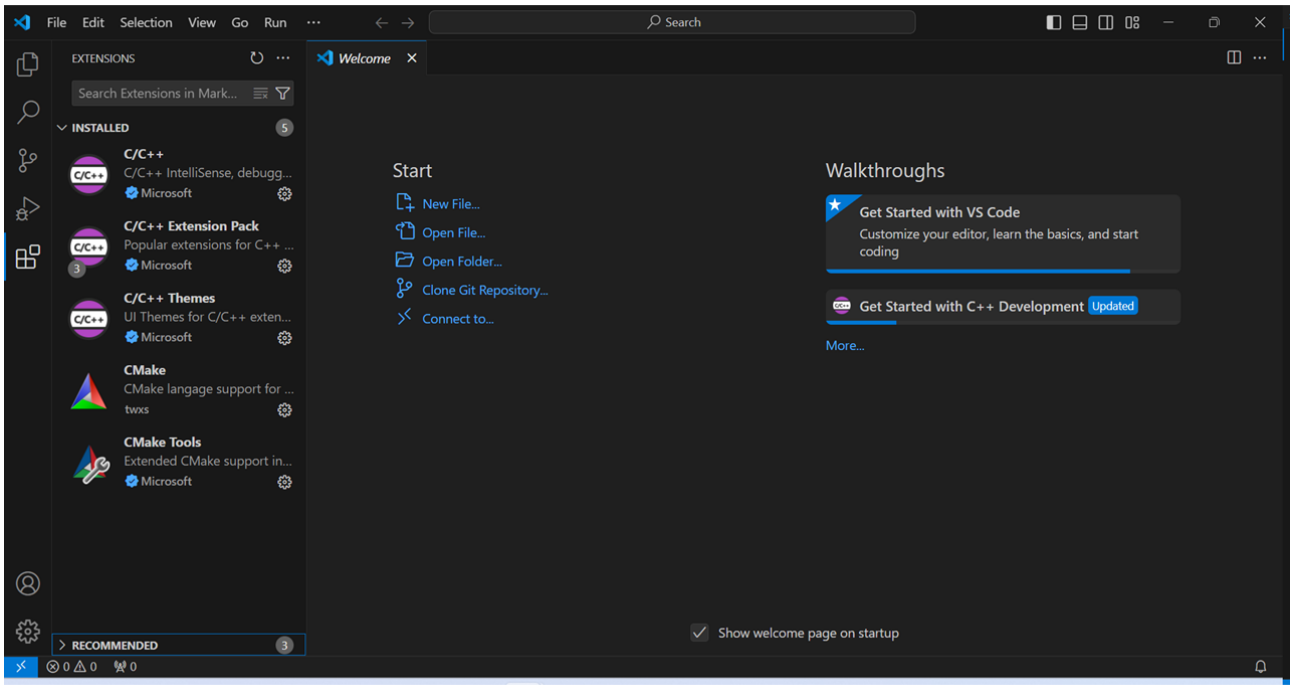
2.2 Installing Extensions

Next, open Visual Studio Code and click on the extensions icon.



Install the following extensions:

- CMake
- CMake Tools
- C/C++
- C/C++ Extension Pack
- C/C++ Themes

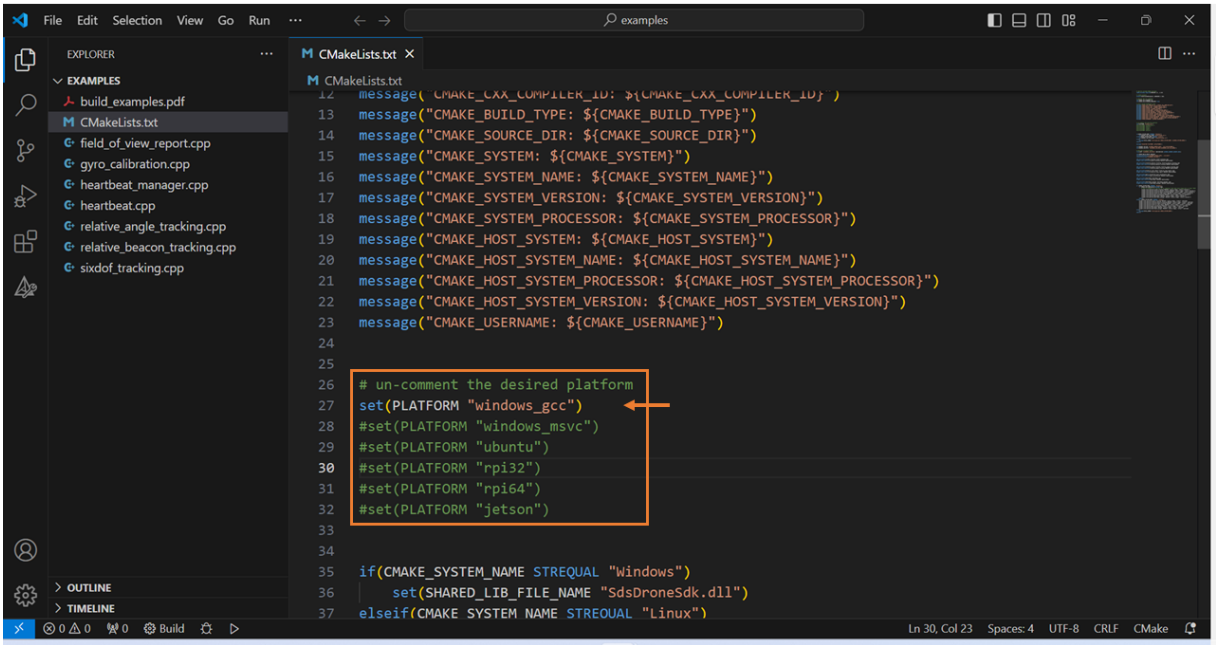

File Edit Selection View Go Run
Search
EXTENSIONS
Search Extensions in Mark...
INSTALLED
C/C++
C/C++ IntelliSense, debugg...
Microsoft
C/C++ Extension Pack
Popular extensions for C++ ...
Microsoft
C/C++ Themes
UI Themes for C/C++ exten...
Microsoft
CMake
CMake langage support for ...
twxs
CMake Tools
Extended CMake support in...
Microsoft
RECOMMENDED
Welcome
Start
New File...
Open File...
Open Folder...
Clone Git Repository...
Connect to...
Walkthroughs
Get Started with VS Code
Customize your editor, learn the basics, and start coding
Get Started with C++ Development
Updated
More...
Show welcome page on startup

3 Running the Examples

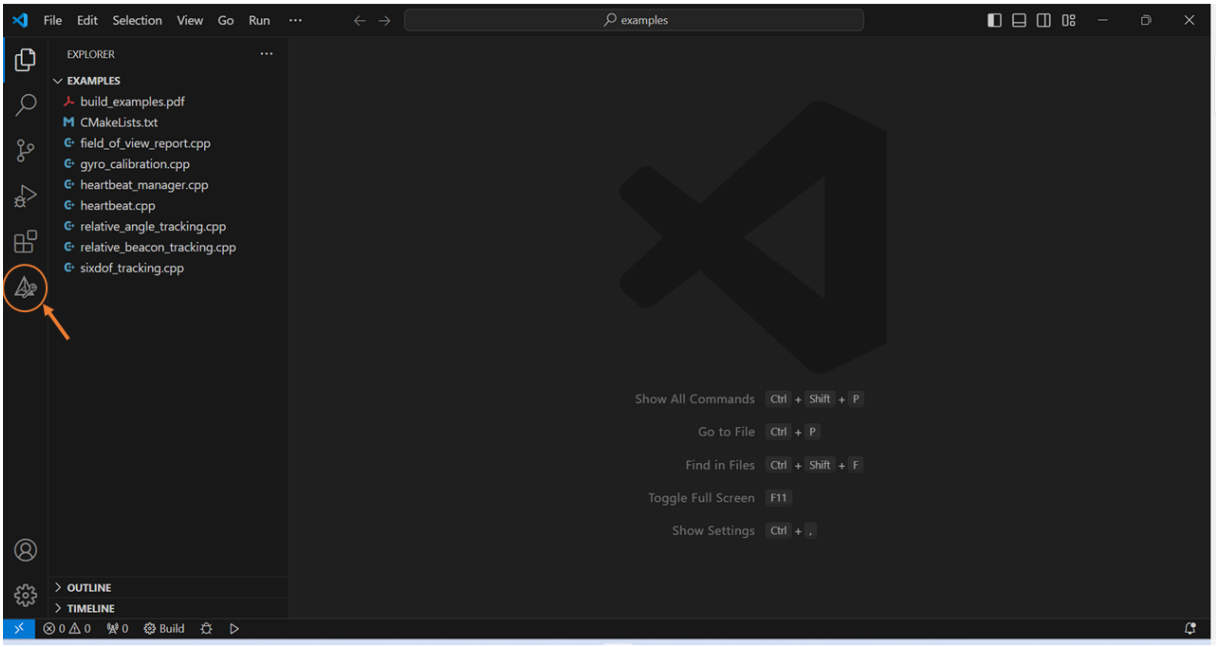
In this section, we will guide you through configuring Visual Studio Code to run the build examples smoothly. Before proceeding, ensure you have a compiler installed on your system and added to your environment variables PATH.

3.1 Project Configuration

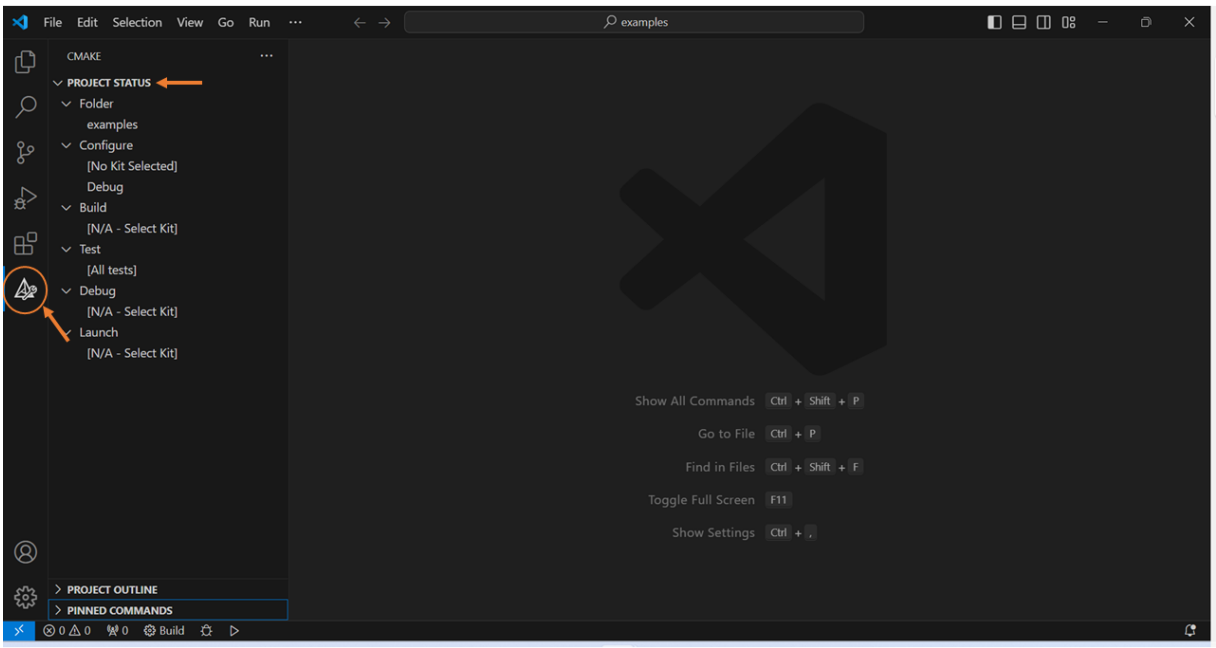
- Open Visual Studio Code in the `examples` folder.
- Review the code to locate where the network configuration details are set. Enter your sensor ID and network name accordingly.
- If you are running the `sixdof_tracking` example or any of the heartbeat examples, ensure you enter your map. Enter each beacon separately in a 4-tuple format: {x, y, z, id}.
- Open the "CMakeLists.txt" file and un-comment the line with the platform you are using.



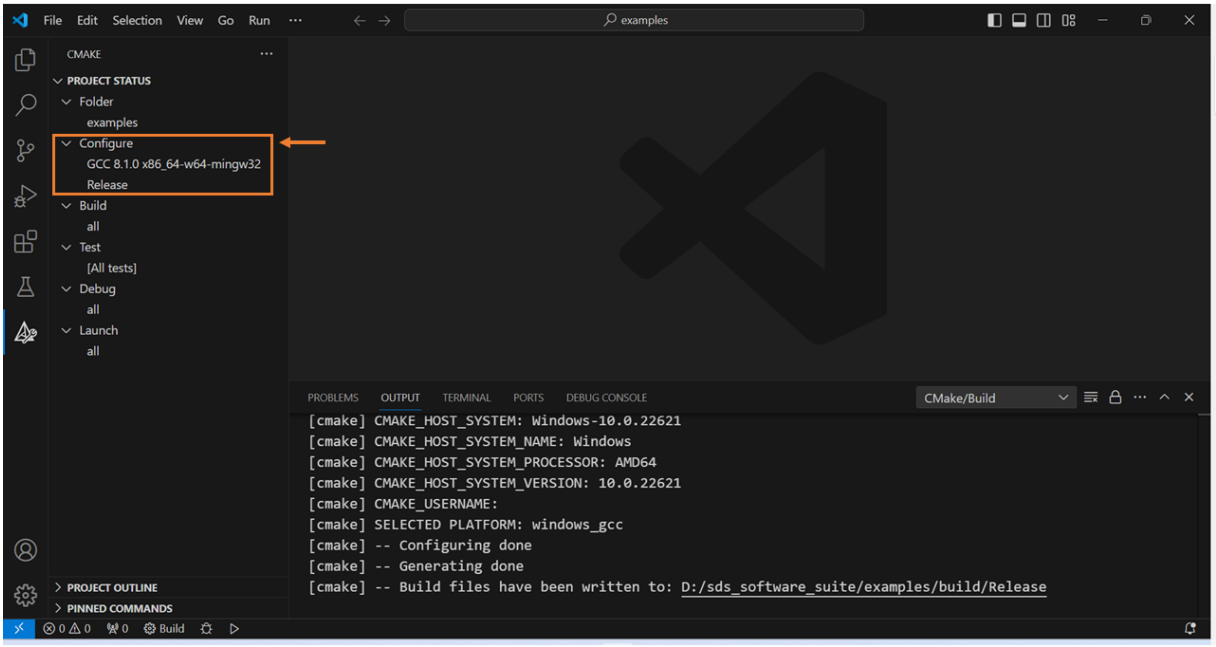
- Locate the CMake icon.



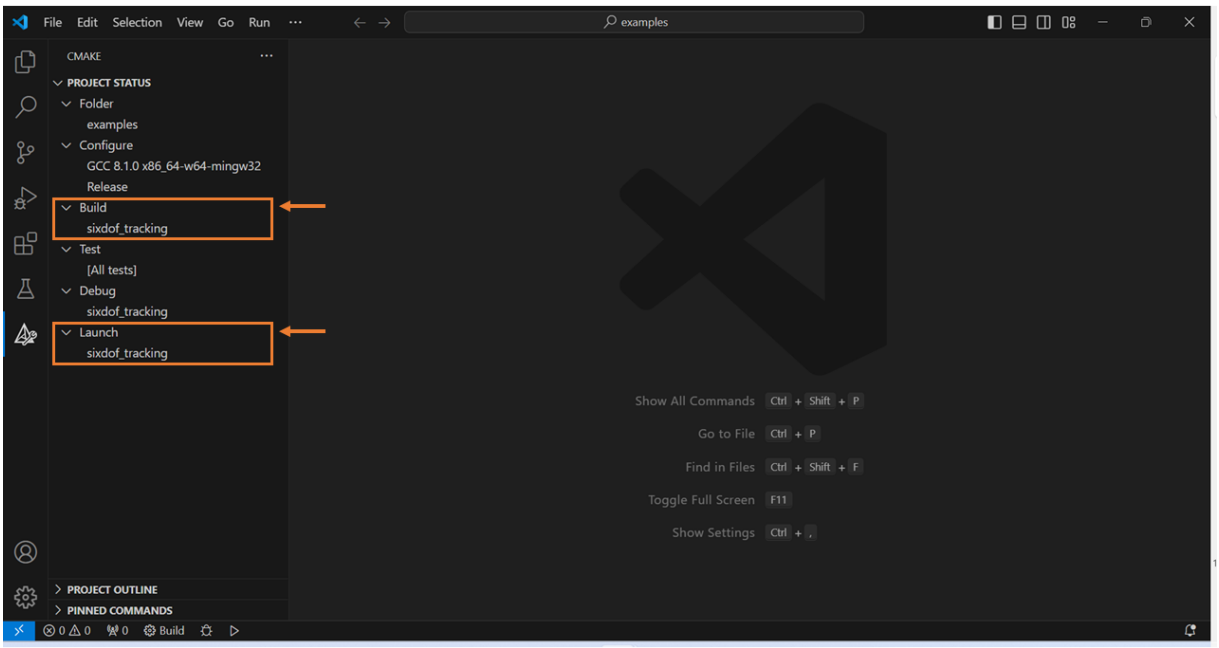
- Click on the CMake icon and check the "Project Status" section.



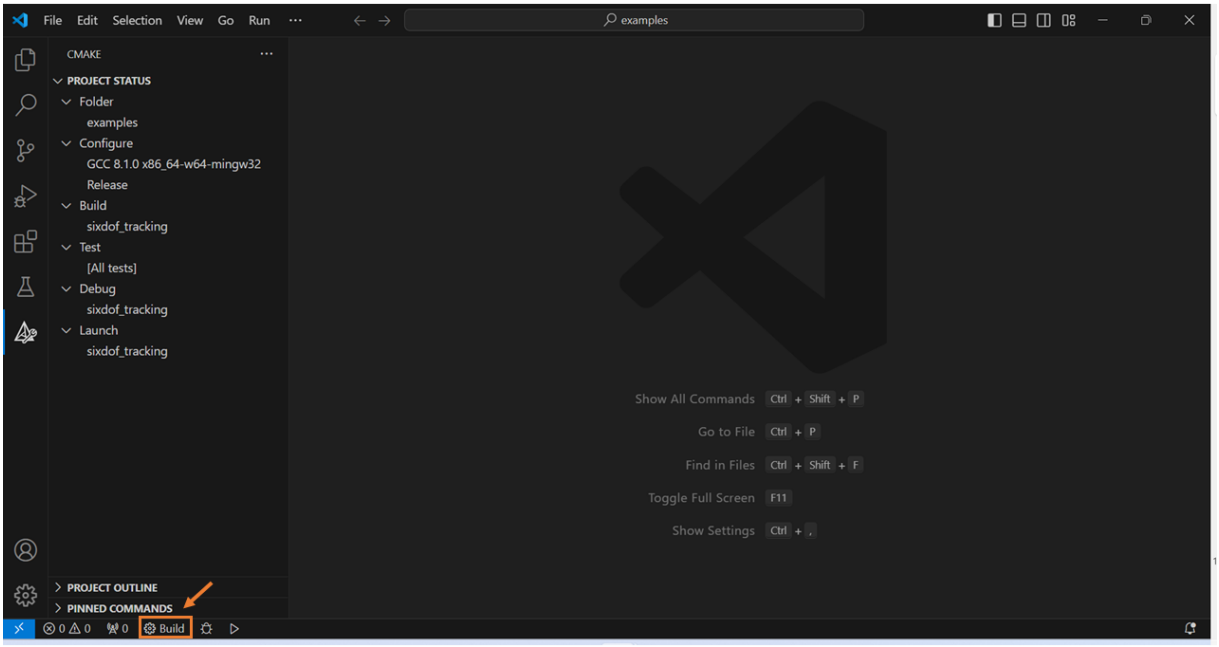
- Under "Configure" select your compiler kit and choose "Release" (not "Debug").



- Under "Build" and "Launch," select the name of the example you wish to build and run. For example, in the image we chose to build `sixdof_tracking`.



- Build the example using the build button located at the bottom left corner.
- Now you should have a `build` folder, containing the executable that you built. Copy the shared library, `SdsFalcon.dll` on Windows or `libSdsFalcon.so` on Linux, from the platform specific folder within the `bin` folder, to the `build` folder.



- Connect the sensor's USB to a 5V power source, and the sensor's RJ45 port to a network port on the host computer or to a USB dongle.
- Set up the beacons in a map. For more information on how to set up a map, see the section "Hardware Setup - Creating a Map" in the Sixdof Falcon User Guide pdf.
- Run the executable file in a terminal opened in `examples\build\Release` (the location of the executable file). Alternatively, you can press the play button on the bottom left of the screen, which will rebuild and then run the executable.

